

Study of Thin Polymer pre-charge Multi point Tactile device

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ABSTRACT

Thin and multi point tactile device is good for human interaction at Metaverse, health, remote control and so on. Pre charge tactile device is very thin, then we can use many applications. We develop the multi point device and confirmed the vibration ability.

Keywords: Tactile device, Polymer type, Thin device, Pre charge.

Index Terms: [Human-centered computing]: Human computer interaction (HCI) - Interaction device – Haptic device

1 INTRODUCTION

Remarkable development of XR technology in recently, flexible piezoelectric tactile devices are required for virtual reality (Metaverse) space, communication robots that coexist with humans and Remote control robot [1]. Since this device can also operate as a sensor, it can exchange tactile sensations with each other and perform tactile communication in both directions in the XR space [2]. It can also be attached on a remote robot hands to present sensing information to the operator on the same device. At this time, the thinness of the device has a feature that it does not interfere with sensing and operation. Recently, a piezoelectric thin film device has been developed in which the upper part and the lower part of silicon rubber are sandwiched between fluorinated ethylene propylene (FEP) and holes are formed inside in it. This device achieves high performance by charging Dozens of kilovolts or more with a corona charge. [3,4], In Reference 2), it was reported that a higher piezoelectric constant could be achieved than ceramic-based piezoelectric material such as PZT, and a prototype was made to confirm its performance. In this report, we prototyped a 0.25mm thin film multi-point tactile device and verified its vibration function. facilisi.

2 FABRICATION METHOD AND DEVICE CONFIGURATION

2.1 Fabrication Method

The prototype of the device was carried out according to the following procedure.

- 1) In order to mold perforated silicone rubber, make a mold with a 3D printer or punched paper patterns. These shape data were designed with 3D and 2D CAD.
- 2) Using the mold and spin coater created a silicon rubber film with a thickness of 0.2mm. EcoFlex 00-30 was used as the silicon rubber. In order to obtain a uniform film thickness, we adopted a method of punching holes with a punch rather than a mold.
- 3) On one side of the FEP film, an aluminum electrode film of about 300 nm was deposited by a resistance heating type evaporator, and a Ti electrode film of about 150 nm was deposited on one side of the multi-point device by sputtering.

- 4) The silicone rubber film and FEP film were treated with oxygen plasma (100 W, 100 Pa, O₂: 30sccm). The treatment was carried out for 6 minutes for silicone rubber and 3 minutes for FEP film. The outline is shown in Fig. 1.
- 5) The silicone resin was soaked in a 10% APTES solution as an adhesive for 24 hours.

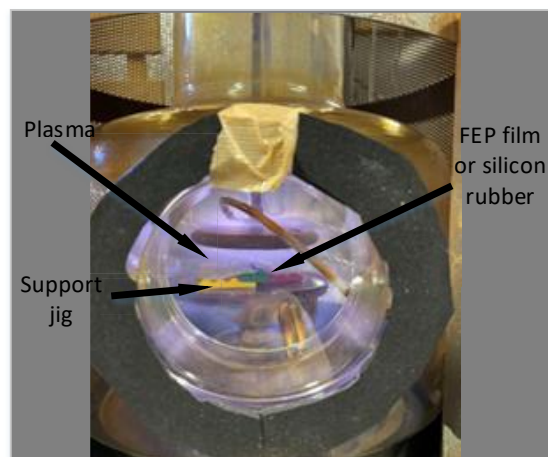


Figure 1: O₂ plasma process.

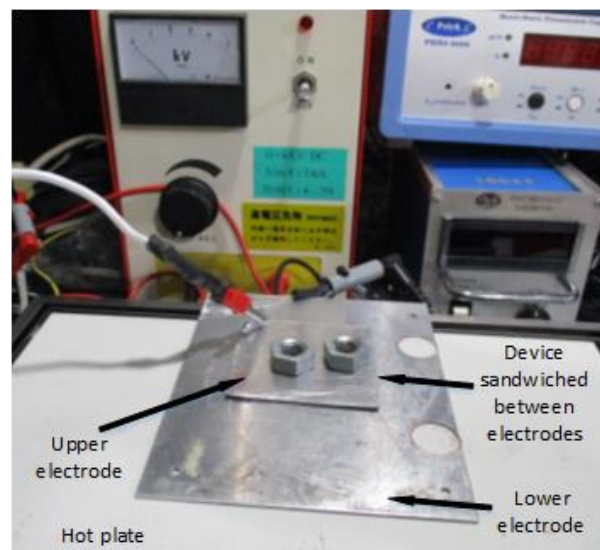


Figure 2: Direct charge process.

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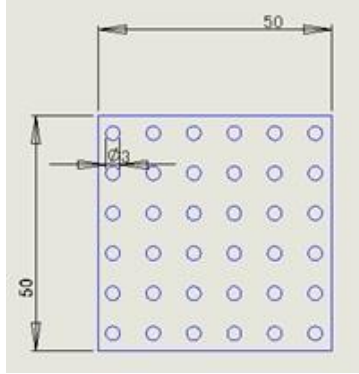


Figure 3: L-1 pattern.

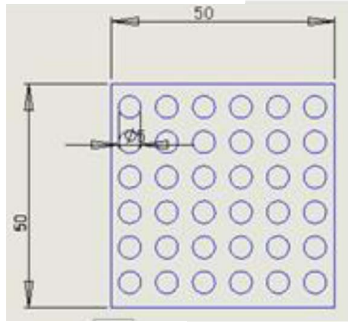


Figure 4: L-2, L-3 pattern.

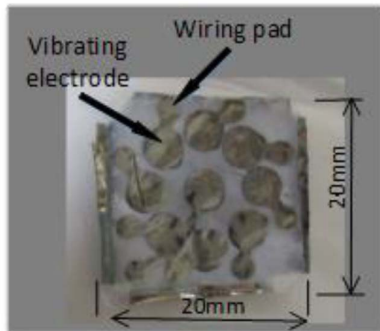


Figure 5: Multi points patterns with contact pad.

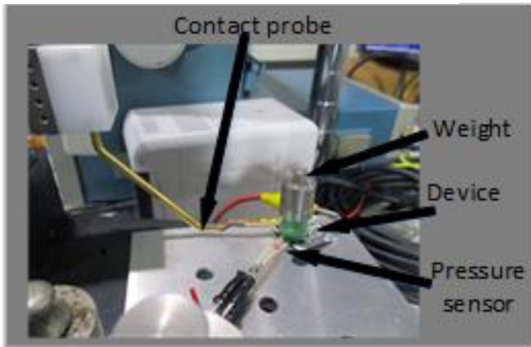


Figure 6: Force measurement method.

- 6) The silicone rubber film and the FEP film were bonded on a hot plate at 40 ° C for 6 hours while being pressed with a weight of 200 g, and a prototype device was produced.
- 7) Corona charge requires a voltage of dozens of KV or more. Since the power supply is also limited, we used direct charging was performed at 1KV for 20 minutes at 120 ° C by sandwiching it directly between electrodes. The outline is shown in Fig. 2.
- 8) It has also been reported that potassium doping to SiO₂ is effective due to the permanent storage of the charge [5]. Therefore, a device (L-3) that is made by doping a potassium in 0.05 mm thin film glass with at 450 ° C for 20 minutes, charging the glass at 300 ° C for 20 minutes at 200 V, and sandwiching it between silicon rubber and the FEP film on one side.

2.2 Device Configuration

In this study, we created two types of devices. One is a uniform large area device, and the other is a small multi-point device.

- 1) Two types of large area devices were created. The L-1 device is shown in Fig. 3. The area ratio of the holes is 10.2 %. The L-2 and L-3 devices are shown in Fig. 4. The area ratio of the holes is 28.2 %. In addition, L-3 is charged with potassium doping the glass. The thickness of each device is 0.25 mm.
- 2) The small multi-point device has nine tactile generation points within a square area measuring 20 mm each side, and the purpose is to generate tactile sensations at each point separately. Fig. 5 is a created device photograph. Small pad is wiring pad. The right below part of the vibrating electrode is a hole, and the area of the hole is 28 %.
- 3) Fig. 6 shows a measurement method. The voltage was applying a 20 V amplitude sine wave centered on + 10 V from the contact probe, placing the Akozon DF9-40 pressure sensor under the device, and sandwiching it with a weight from above. Large area devices weighed 150 g and multipoint devices weighed 15 g.

3 DEVICE PERFORMANCE

The vibration characteristics of each device were evaluated by the measurement method in the previous section. For the vibration frequency, the vibration force generated from 70 Hz to 300 Hz was measured. These vibration frequencies can be perceived by humans. Table 1 shows the charge amount (voltage) change. The charge amount did not change so much in 3 weeks.

Table 1: Charge voltage change.

	Charge time [V]	3 weeks later [V]
L-1	711	723
L-2	735	716
L-3	694	724
Multi point	138	95

3.1 Large Area Device

Fig. 7 shows the results of measuring the vibration force of the large area device immediately after charging. Vibration was measured from 70Hz to 300Hz. Fig. 8 shows the vibration force measurement results of each device 3 weeks after charging. The L-1 device can generate a large vibration force above 100Hz. It can also be seen

that the L-2 device vibrates stably from 70 Hz. The L-3 (glass was doped with potassium) had the same vibration output as a normal L-2 device. In addition, the L-1 device was observed to vibrate five times as much as the L-2 device. The three types of devices were able to feel sufficient vibration with the fingers.

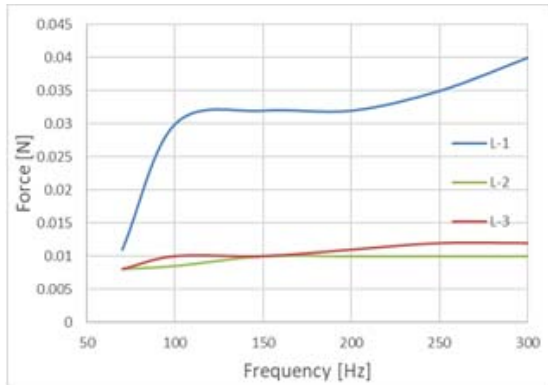


Figure 7: Vibration force (Charge time).

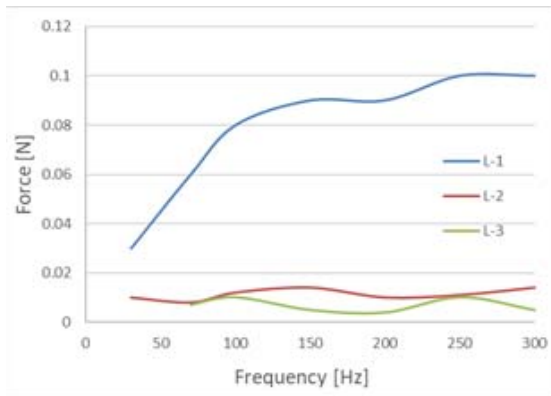


Figure 8: Vibration force (3 week later).

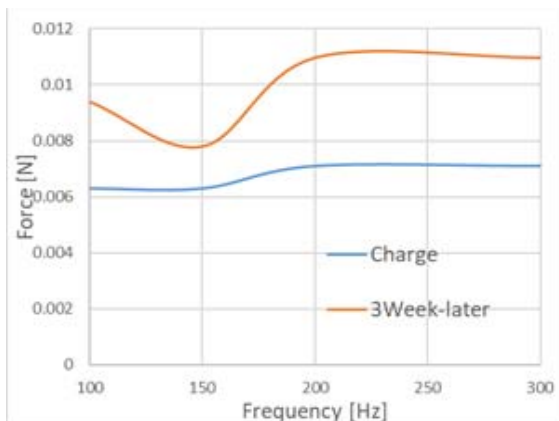


Figure 9: Vibration force of Multi point device.

3.2 Multipoint Device

Fig. 9 shows the measurement result of the vibration force of the small multi-point device. The vibration characteristics immediately after charging and after 3 weeks did not change so much. This device had a lower vibration output than the large device, but we could feel the vibration with fingers similar as lightly moving across sandpaper.

4 CONCLUSION

In this research, we have developed a charge-type polymer thin-film piezoelectric tactile device. First, a large square area measuring 50 mm each side device, the hole area was adjusted to improve the five times higher performance. By sandwiching a potassium-doped thin-film glass plate, which has a very long charge disappearance time, we confirmed similar vibration performance with normal device. In addition, a small nine-point tactile device of square area measuring 20 mm each side can be developed, which is effective for local tactile sensation generation such as fingers. Therefore, we would like to consider the exchange tactile sensations with each other and perform tactile communication in both directions in the XR application.

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